

Name _____

Quiz 2

1. (6 points) Suppose you have algorithms with the following runtimes. (Assume these are exact running times as a function of the input size n , not asymptotic running times.) How much slower do these algorithms get when you double the input size? (You can find this by dividing the runtime on $2n$ by the runtime on n). You should simplify your answer as much as possible, including evaluating logarithms when possible.

(a) $5n^3$

(b) $2n \log_2 n$

(c) 3^n

2. (4 points) Simplify the following Big-O notation so that $f(n) = O(g(n))$. The function $g(n)$ should be both as simple and as tight as possible. For example, if $f(n) = 3n^2 + 5n + 2$, then the answer should be $O(n^2)$, even though $O(n^3)$ is also correct but not as tight, and $O(n^2 + 2)$ is also correct but not fully simplified.

(a) $f(n) = \log(6 \cdot n^{n+1})$. What is $g(n)$?